Problem solving with algebra

1 Adam sells chocolates to raise money for charity. Each chocolate costs $\$6$.

- **a** If Adam sells q chocolates, write an algebraic expression for the amount of money he raises.
- **b** How much do 4 chocolates cost?
- **c** How much do 5 chocolates cost?
- **d** Is it possible to raise exactly $\$13$ if Adam sells chocolates at $\$6$?

2 Valentina's hens can produce 5 eggs each day.

- **a** If Valentina collects the eggs from her hens for y days, write an algebraic expression for the total number of eggs.
- **b** Find the number of eggs Valentina will have after 30 days.

3 John breeds dogs for sale. Each of his d dogs gives birth to p puppies.

- **a** Write an algebraic expression for the total number of puppies.
- **b** Can the value of p be a fraction?
- **c** If he has 3 dogs, find the total number of puppies.

4 Judy has 72 pencils, which she shares evenly among the students in her class.

- **a** If there are q students in her class, write an algebraic expression for the number of pencils given to each student.
- **b** If there are 18 students in the class, how many pencils does each student receive?
- **c** If there are 36 students in the class, how many pencils does each student receive?
- **d** As the number of students in the class increases, does the number of pencils each student receives increase or decrease?

5 At John's birthday party, cake slices are shared evenly among his 8 guests.

- **a** If John shares x slices, write an algebraic expression for the number of slices given to each guest.
- **b** If John shares 32 cake slices, find the number of slices given to each guest.

6 Kenneth uses a watering can to give each of his plants an equal amount of water.

- **a** If his watering can contains m mL of water and he has n plants, write an algebraic expression for how much water is given to each plant.
- **b** As n increases, do the plants receive more or less water?

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13 Vanessa has $\$700$ in her bank account. She only uses the account to pay her mobile phone bill each month.

a If each monthly bill is $\$14$, write an algebraic expression for how much Vanessa has in her account after c months.

b Find the amount of money she will have in her account after she pays her bill for 4 months.

14 Sally attends a dessert festival. Each ice cream costs $\$6$ and each milkshake costs $\$8$.

a If Sally buys c ice creams and m milkshakes, write an algebraic expression for the total cost.

b How much would two ice creams and one milkshake cost?

c Using algebra, how much would c ice creams and no milkshakes cost?

15 Laura has a piggy bank in which she collects $20c$ and $50c$ coins. After some time, she loses track of how many coins are in the piggy bank.

a Let m represent the number of $20c$ coins and n represent the number of $50c$ coins. Write an algebraic expression for the total value of Laura's coins in cents.

b Laura breaks her piggy bank and discovers that she has thirteen $20c$ and twenty-seven $50c$ coins. Find the total value of these coins in dollars.

16 Valerie places a bird feeder in her garden. That day, she sees 3 birds use the feeder. The next day she sees 6 birds, and on the third day she sees 9 birds.

a If the number of birds continues to follow the pattern, find the number of birds that Valerie will see on the fourth day.

b If the pattern continues, write an algebraic expression for the number of birds Valerie sees on the x th day.

17 Water is dripping from a tap into a large bucket, so that:

- After 1 hour, the water level in the bucket is 5 cm
- After 2 hours the water level is at 10 cm
- After 3 hours the water level reaches 15 cm

a If the tap stops leaking after a hours, write an algebraic expression for the water level in the bucket at this time.

b Is the water level increasing by the same amount each hour?

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18 Fred likes to go kayaking. He takes his boat down to the lake which is 400 m away from his house.

- After 1 minute of paddling he is 490 m away from his house
- After 2 minutes of paddling he is 580 m away from his house
- After 3 minutes of paddling he is 670 m away from his house

a Fred paddles for n minutes in total. Write an algebraic expression for the distance from his house at this time.

b How far does Fred travel in the kayak per minute?

c Is Fred is paddling at a constant pace for the first n minutes?

d If Fred kayaks for 10 minutes, how far does he travel from his house altogether?

19 Kathleen cuts squares from 1 cm grid paper:

- The first square is $2\text{ cm} \times 2$ cm and has an area that contains four 1 cm^2 pieces
- The second square is $3\text{ cm} \times 3$ cm and has an area that contains nine 1 cm^2 pieces
- The third square is $4\text{ cm} \times 4$ cm and has an area that contains sixteen 1 cm^2 pieces

a If the pattern continues, find the number of 1 cm^2 pieces that fit in a square that is $5\text{ cm} \times 5$ cm.

b Write an algebraic expression for the number of 1 cm^2 are in a square with a side length of h cm.

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